ECON 6018
TUTORIAL 6

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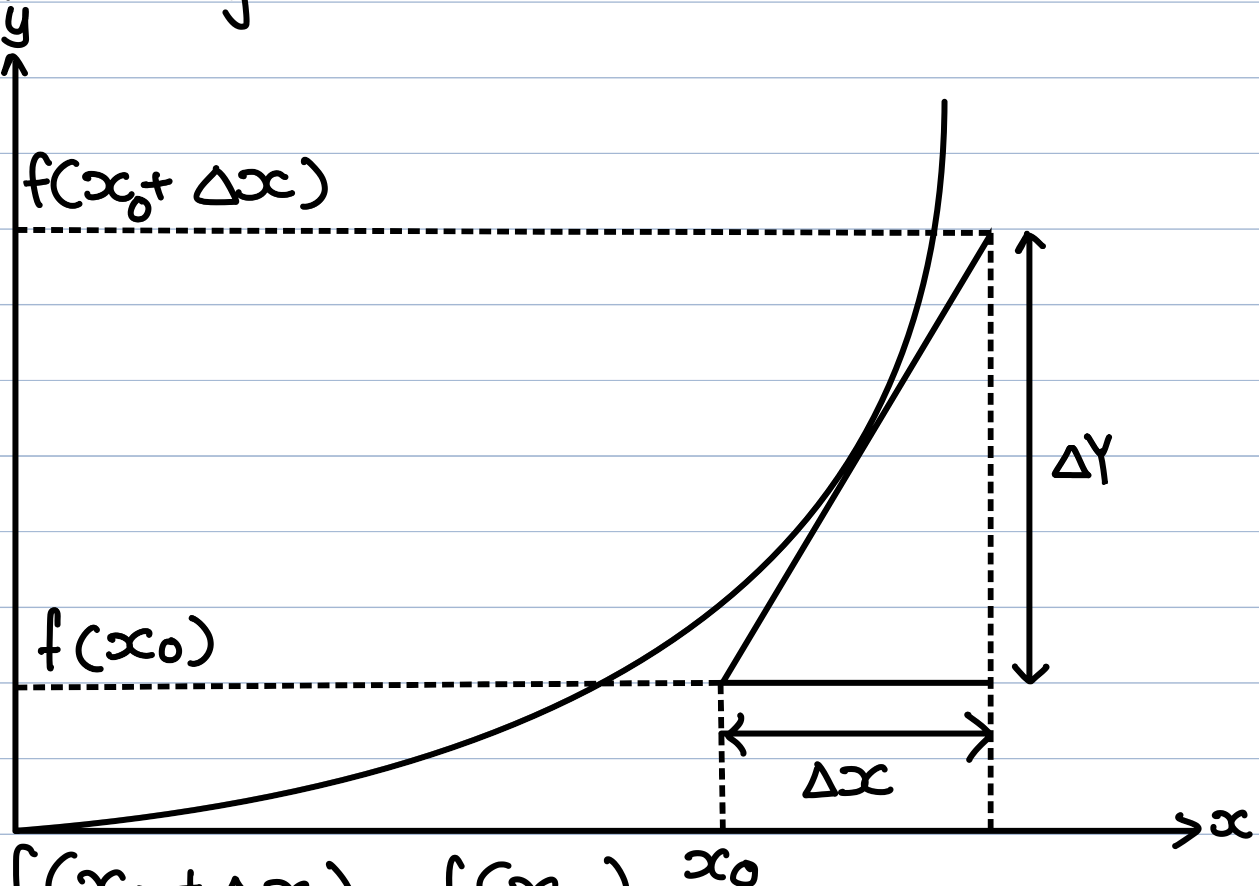
1. What are Differentials?

Previously,
we learnt $\frac{dy}{dx}$ (Derivatives)

Now,
- focus on individual dy and dx
- Differentials: Δy and Δx

Differential —— small changes

Graph —— $y = 2x^2$



$$\Delta y = f(x_0 + \Delta x) - f(x_0)$$
$$= (x_0 + \Delta x)^2 - x_0^2 = x_0^2 + 2x_0\Delta x + \Delta x^2 - x_0^2$$
$$= 2x_0\Delta x + \Delta x^2$$

2. Differential of X & Y

First formula:

1. $dy = f'(x)dx$

Practices:

1. $y = x^2 + 2$

2. $y = \ln(x^2 + 2x)$

3. $z = \dfrac{x^3}{(x^2+3)^2}$

4. $z = e^{2x^2+x}$

Answers:

1. $y = x^2 + 2$

$dy = dx^2 + d2$

$dy = 2x\,dx$

2. $y = \ln(x^2 + 2x)$

$dy = d\ln(x^2 + 2x)$

$dy = \dfrac{2x + 2}{x^2 + 2x}\,dx$

3. $z = \dfrac{x^3}{(x^2+3)^2}$

$dz = d\dfrac{x^3}{(x^2+3)}$

$dz = \left[\dfrac{3x^2 \cdot (x^2+8) - x^3(2x)}{(x^2+3)^2}\right] dx$

4. $z = e^{2x^2+x}$

$dz = de^{2x^2+x}$

$dz = (4x+1)e^{2x^2+x}dx$

First formula:

1. $dy = f'(x)dx$

Abstract Qns:

1. Easy $y = ax^b + c$, where a, b, c are numerical constant

2. Normal $y = \alpha \ln(\beta x^3 + x)^2$ where α, β are numerical constants

3. Hard

$y = \frac{(ax+b)^m}{(cx+d)^n}$ where a, b, c, d, m, n are constant $ad - bc \neq 0$

Answers:

1. $y = ax^b + c$

$dy = dax^b + dc$

$dy = adx^b$

$dy = abx^{b-1}dx$

2. $y = \alpha \ln(\beta x^3 + x)^2$

$y = 2\alpha \ln(\beta x^3 + x)$

$dy = d\,2\alpha \ln(\beta x^3 + x)$

$dy = 2\alpha\, d\ln(\beta x^3 + x)$

$dy = 2\alpha\left(\dfrac{3\beta x^2 + 1}{\beta x^3 + x}\right)dx$

3. $y = \dfrac{(ax+b)^m}{(cx+d)^n}$

$$\frac{dy}{dx} = \frac{d}{dx}\frac{(ax+b)^m}{(cx+d)^n}$$

$$\frac{dy}{dx} = \frac{m(ax+b)^{m-1}(a).(cx+d)^n - (ax+b)^m . n(cx+d)^{n-1}.c}{[(cx+d)^n]^2}$$

3. Total Differentiation

- formula for multivariate functions

Second Formula:

1. $f(x, y)$

$$df = f'_x \, dx + f'_y \, dy$$

Practices

1. $z = x^2y + xy^3$

2. $z = \dfrac{x^2 + y^2}{xy}$

Answer:

1. $z = x^2y + xy^3$ (Product)

$$dz = dx^2y + dxy^3$$

$$dz = x^2dy + ydx^2 + x\,dy^3 + y^3dx$$

$$= x^2dy + 2xydx + 3y^2xdy + y^3dx$$

$$= (x^2 + 3xy^2)dy + (2xy + y^3)dx$$

2. $z = \dfrac{x^2 + y^2}{xy}$ (Quotient)

$$dz = d\,\frac{x^2 + y^2}{xy}$$

$$dz = \frac{(xy)\,d(x^2 + y)^2 - (x^2 + y^2)\,d\,(xy)}{(xy)^2}$$

$$= \frac{(xy)\,(2xdx + 2ydy) - (x^2 + y^2)\,[ydx + xdy)}{(xy)^2}$$

Abstract Qns:

1. $z = (ax+b)^m (cy+d)^n$

- where a, b, m, c, d, n are numerical constants

2. HARD

$$z = \frac{(ax+by)^m}{(cx+dy)^n}$$

- where a, b, m, c, d, n are numerical constants

$cx+dy \neq 0$

Ans:

1. $z = (ax+b)^m (cy+d)^n$

$dz = d(ax+b)^m (cy+d)^n$

$dz = (cy+d)^n d(ax+b)^m + (ax+b)^m d(cy+d)^n$

$dz = (cy+d)^n \cdot m(ax+b)^{m-1} \cdot a dx$

$+ (ax+b)^m \cdot n(cy+d)^{n-1} \cdot c dy$

2. $z = \dfrac{(ax+by)^m}{(cx+dy)^n}$

$dz = d\dfrac{(ax+by)^m}{(cx+dy)^n}$

$dz = \dfrac{(cx+dy)^n d(ax+by)^m - (ax+by)^m d(cx+dy)^n}{[(cx+dy)^n]^2}$

$$d(ax+by)^m = m(ax+by)^{m-1}\,d(ax+by) \quad \text{Chain Rule}$$

$$= m(ax+by)^{m-1}(a\,dx + b\,dy)$$

$$d(cx+dy)^n = n(cx+dy)^{n-1}\,d(cx+dy)$$

$$= n(cx+dy)^{n-1}(c\,dx + d\,dy)$$

$$dz = \frac{(cx+dy)^n\, m(ax+by)^{m-1}(a\,dx+b\,dy) - (ax+by)^m\, n(cx+dy)^{n-1}(c\,dx+d\,dy)}{\left[(cx+dy)^n\right]^2}$$

Application Qns

1. $Y = C + I + G + \frac{X}{M}$

2. $Y = C(I) + \frac{GT}{L}$

Ans:

$$Y = C + I + G + \frac{X}{M}$$

$$dY = dC + dI + dG + \frac{dX}{M}$$

$$= dC + dI + dG + \frac{MdX - XdM}{M^2}$$

$$Y = C(I) + \frac{GT}{L}$$

$$dY = d\,C(I) + d\frac{GT}{L}$$

$$= C'(I)dI + \frac{LdGT - GTdL}{L^2}$$

$$= C'(I)dI + \frac{L(TdG + GdT) - GTdL}{L^2}$$